



2017 HSC Course Assessment Task 4 (Trial HSC)

Thursday August 3, 2017

SECTION I

- Mark your answers on the answer grid provided (on page 13)

SECTION II

- Commence each new question on a new booklet. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

Class (please ✓)

○ 12MAT.2 – Mr Lam

Marker's use only.

[illegible]

Section I

10 marks

Attempt Question 1 to 10

Allow approximately 10 minutes for this section

Mark your answers on the answer grid provided (labelled as page 13).

Glossary

- $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3\}$ – set of all integers.
- $\mathbb{Z}^+$ – all positive integers (excludes zero)
- $\mathbb{R}$ – set of all real numbers
- $\mathbb{C}$ – set of all complex numbers

Questions

Marks

1. What is the number of asymptotes on the graphs of $y = \frac{x^3}{x^2 - 1}$? 1
- (A) 1 (B) 2 (C) 3 (D) 4

2. The equation $x^3 - y^3 + 3xy + 1 = 0$ is an implicit function in x and y . 1

Which of the following is the expression for the gradient function?

- (A) $\frac{y^2 - x}{x^2 + y}$ (B) $\frac{y^2 + x}{x^2 - x}$ (C) $\frac{x^2 + y}{y^2 - x}$ (D) $\frac{x^2 - y}{y^2 + x}$

3. The polynomial $P(z) = z^4 - 4z^3 + Az + 20$, where $A \in \mathbb{R}$, has $(3 + i)$ as one of its zeros. 1

Which of the following expression is $P(z)$ expressed as a product of two real quadratic factors?

- (A) $(z^2 - 2z + 2)(z^2 - 6z + 10)$ (C) $(z^2 - 2z + 2)(z^2 + 6z + 10)$
- (B) $(z^2 + 2z + 2)(z^2 - 6z + 10)$ (D) $(z^2 + 2z + 2)(z^2 + 6z + 10)$

4. A particle of mass m is moving horizontally in a straight line. It experiences a resistive force of magnitude $2m(v + v^2)$ N when its speed is v metres per second. 1

At time t seconds, the particle has a displacement of x metres from a fixed point O .

Which of the following is the correct expression for x in terms of v ?

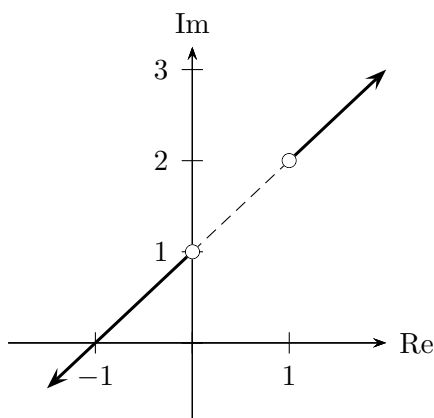
- (A) $x = -\frac{1}{2} \int \frac{1}{1+v} dv$ (C) $x = \frac{1}{2} \int \frac{1}{1+v} dv$
 (B) $x = -\frac{1}{2} \int \frac{1}{v(1+v)} dv$ (D) $x = \frac{1}{2} \int \frac{1}{v(1+v)} dv$

5. Which of the following could be $f(x)$ if 1

$$\int f(x) \sin x \, dx = -f(x) \cos x + \int 3x^2 \cos x \, dx$$

- (A) $3x^2$ (B) x^3 (C) $-x^3$ (D) $-3x^2$

6. Which of the following defines the locus of the complex number z , as sketched below? 1



- (A) $\arg\left(\frac{z-i}{z-1-2i}\right) = \pi$ (C) $\arg\left(\frac{z+i}{z-1-2i}\right) = \pi$
 (B) $\arg(z+i) = \arg(z-1-2i)$ (D) $\arg(z-i) = \arg(z-1-2i)$

7. Multiplying a non-zero complex number by $\frac{\frac{1}{2} - \frac{\sqrt{3}}{2}i}{\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i}$ results in a rotation about the origin on an Argand diagram. 1

What is the rotation?

- (A) Clockwise by $\frac{\pi}{12}$ (C) Anticlockwise by $\frac{\pi}{12}$
 (B) Clockwise by $\frac{7\pi}{12}$ (D) Anticlockwise by $\frac{7\pi}{12}$

8. Without evaluating the integrals, which of the following is greater than zero? 1

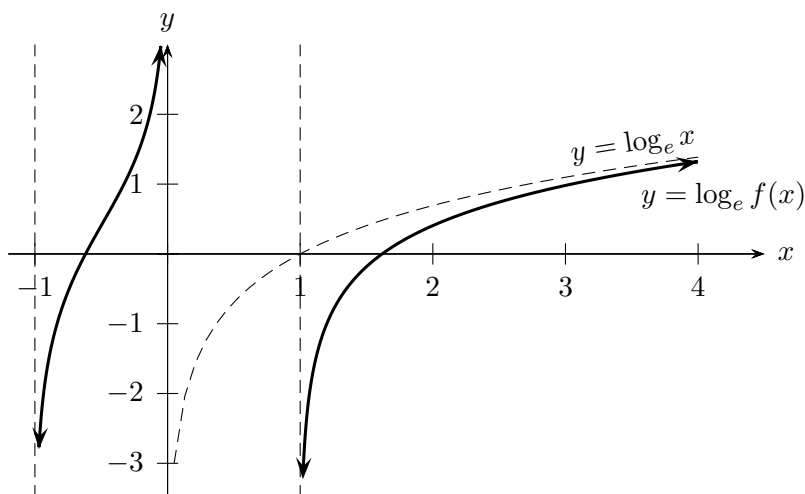
(A) $\int_{-1}^1 \tan^{-1}(\sin x) \, dx$

(C) $\int_{-1}^1 \left((e^x)^3 + x^7 \right) \, dx$

(B) $\int_{-1}^1 \frac{2x}{\sin^2 x} \, dx$

(D) $\int_{-1}^1 \frac{x^5}{\cos^3 x} \, dx$

9. The following is a graph of $y = \log_e f(x)$ over its natural domain. The graph $y = \log_e f(x)$ has vertical asymptotes at $x = 0$, $x = \pm 1$, and an upper bound of $y = \log_e x$ as $x \rightarrow \infty$. 1



Which of the following is the correct expression for $f(x)$?

(A) $y = x - \frac{1}{x}$

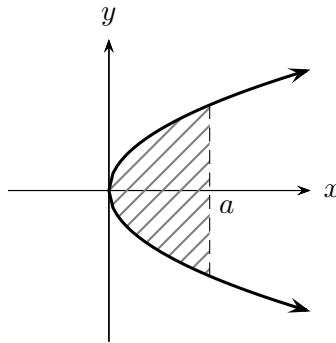
(C) $y = -\frac{1}{x}$

(B) $y = x + \frac{1}{x}$

(D) $y = \frac{1}{x}$

10. A solid is formed by rotating the region enclosed by the parabola $y^2 = 4ax$, its vertex $(0, 0)$ and the line $x = a$ about the y axis.

1



Which of the following integrals gives the volume of this area by *slicing*?

(A) $2\pi\sqrt{a} \int_0^a z^{\frac{3}{2}} dz$

(C) $\pi \int_0^{2a} \left(a^2 - \frac{z^4}{16a^2} \right) dz$

(B) $4\pi\sqrt{a} \int_0^a z^{\frac{3}{2}} dz$

(D) $2\pi \int_0^{2a} \left(a^2 - \frac{z^4}{16a^2} \right) dz$

Examination continues overleaf...

Section II

90 marks

Attempt Questions 11 to 16

Allow approximately 2 hours and 50 minutes for this section.

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 Marks)		Commence a NEW booklet.	Marks
(a)	Find $\int e^{-2x} \cos x \, dx$.		4
(b)	Evaluate $\int \frac{x}{\sqrt{1-x}} \, dx$.		3
(c)	i. Using the substitution $x = a - u$, show that $\int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx$.		2
	ii. Hence evaluate $\int_0^1 x(1-x)^{2017} \, dx$, giving your answer as the simplest fraction.		2
(d)	Use the substitution $t = \tan \frac{x}{2}$ to evaluate		4
	$\int_0^{\frac{\pi}{2}} \frac{1}{3 - \cos x - 2 \sin x} \, dx$		

Examination continues overleaf...

- | | | |
|-------------------------------|-------------------------|--------------|
| Question 12 (15 Marks) | Commence a NEW booklet. | Marks |
|-------------------------------|-------------------------|--------------|
- (a) i. Express $z = 1 + \sqrt{3}i$ in modulus-argument form. **1**
- ii. Hence or otherwise, show that $z^7 - 64z = 0$. **2**
- (b) Find $\sqrt{6i - 8}$, and hence solve the equation **4**
- $$2z^2 - (3 + i)z + 2 = 0$$
- (c) Find the complex number $z = a + ib$, where $a, b \in \mathbb{R}$, such that $2\bar{z} - iz = 1 + 4i$. **2**
- (d) Sketch the region in the Argand diagram of the point z such that it satisfies all of **3**
- $$\begin{cases} |\text{Arg}(z)| < \frac{\pi}{3} \\ z + \bar{z} < 4 \\ |z| > 2 \end{cases}$$
- (e) $ABCD$ is a quadrilateral in the complex plane such that the vectors $\overrightarrow{OA}$, $\overrightarrow{OB}$, $\overrightarrow{OC}$ and $\overrightarrow{OD}$ represent complex numbers a , b , c and d respectively.
- P , Q , R and S are the midpoints of AB , BC , CD and DA respectively. M and N are midpoints of PR and QS respectively.
- i. Show that the vectors $\overrightarrow{OM}$ and $\overrightarrow{ON}$ both represent the complex number **2**
- $$\frac{1}{4}(a + b + c + d)$$
- ii. Hence explain the type of quadrilateral that $PQRS$ is. **1**

Examination continues overleaf...

Question 13 (15 Marks) Commence a NEW booklet. **Marks**

- (a) α , β and γ are non-zero and the roots of the cubic equation

$$x^3 + px + q = 0$$

- i. Express $\alpha^2 + \beta^2 + \gamma^2$ in terms of p and q . **1**
- ii. Hence or otherwise, form a cubic equation with roots **3**

$$\frac{\alpha}{\beta\gamma}, \frac{\beta}{\alpha\gamma} \text{ and } \frac{\gamma}{\alpha\beta}$$

- (b) i. Find A , B and $C \in \mathbb{R}$ such that **3**

$$\frac{4x^2 - 5x - 7}{(x-1)(x^2 + x + 2)} \equiv \frac{A}{x-1} + \frac{Bx + C}{x^2 + x + 2}$$

- ii. Hence evaluate **2**

$$\int_{-1}^0 \frac{4x^2 - 5x - 7}{(x-1)(x^2 + x + 2)} dx$$

- (c) i. Prove for all A : **2**

$$\cos^3 A - \frac{3}{4} \cos A = \frac{1}{4} \cos 3A$$

- ii. Show that $x = 2\sqrt{2} \cos A$ satisfies the cubic equation **2**

$$x^3 - 6x = -2$$

$$\text{provided } \cos 3A = -\frac{1}{2\sqrt{2}}.$$

- iii. Hence or otherwise, find all three roots to the equation $x^3 - 6x + 2 = 0$, **2**
correct to four decimal places.

Examination continues overleaf...

Question 14 (15 Marks)

Commence a NEW booklet.

Marks

- (a) i. Determine the real values of p for which the equation

$$\frac{x^2}{3+p} + \frac{y^2}{8+p} = 1$$

defines

(α) an ellipse **1**

(β) a hyperbola **2**

- ii. For the value $p = -4$ in the above equation, find the **2**

- eccentricity
- coordinates of the foci, and
- the equations of the directrices

of the conic.

- iii. Draw a neat sketch of the conic in part (ii), indicating the foci, vertices and directrices. **2**

- (b) i. Prove that the equation of the tangent at the point $\left(t, \frac{1}{t}\right)$ to the hyperbola **2**

$$xy = 1 \text{ is } x + t^2y = 2t.$$

- ii. The tangent at a point P on the hyperbola $xy = 1$ meets the y axis at A , **3**
and the normal at P meets the x axis at B .

Show that the equation of the locus of the midpoint of AB as P moves along the hyperbola is

$$x = \frac{1 - y^4}{2y}$$

- (c) $P(a \cos \alpha, b \sin \alpha)$ and $Q(a \cos \beta, b \sin \beta)$ are the endpoints of a focal chord of **3**
the curve $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Show that $e = \frac{\sin(\alpha - \beta)}{\sin \alpha - \sin \beta}$.

Examination continues overleaf...

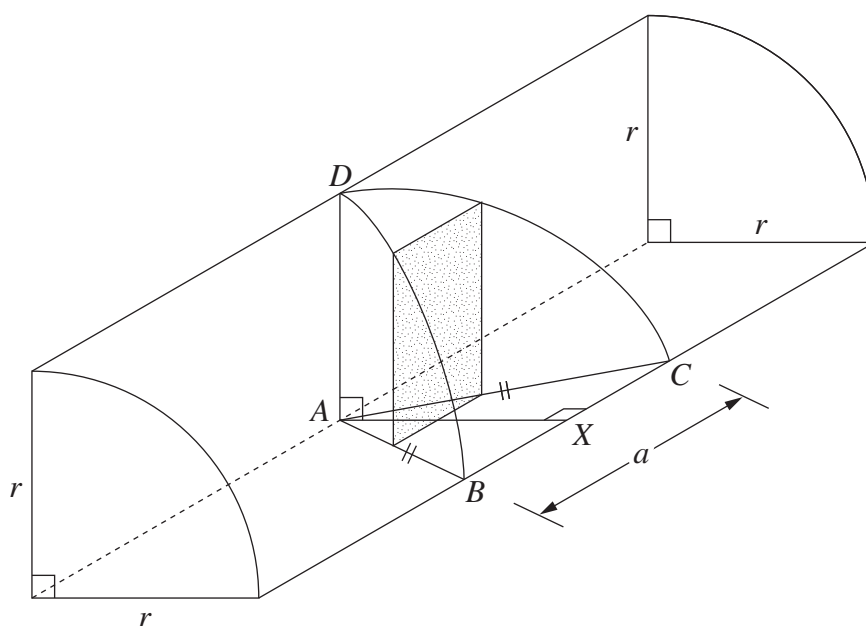
Question 15 (15 Marks)

Commence a NEW booklet.

Marks

- (a) By using the method of cylindrical shells, find the volume of the solid generated when the region bounded by the curve $y = \log_e x$, the x axis and the lines $x = 1$ and $x = e$ is rotated about the y axis. **3**
- (b) The solid $ABCD$ is cut from a quarter cylinder of radius r as shown. Its base is an isosceles $\triangle ABC$ with $AB = AC$. The length of BC is a and the midpoint of BC is X . **3**

The cross-sections perpendicular to AX are rectangles. A typical cross-section is shown shaded in the diagram.



Find the volume of the solid $ABCD$.

- (c) A particle of unit mass moves in a straight line and experiences a resistive force of $v + v^3$, where v is its velocity. **2**

Initially, the particle is at the origin and travelling to the right with speed Q metres per second.

- i. Show that the displacement of the particle is given by **2**

$$x = \tan^{-1} \left(\frac{Q - v}{1 + Qv} \right)$$

- ii. Show that at time t which has elapsed, the velocity-time relationship is given by **3**

$$t = \frac{1}{2} \log_e \left(\frac{Q^2 (1 + v^2)}{v^2 (1 + Q^2)} \right)$$

- iii. Find v^2 as a function of time. **2**
- iv. Find the limiting values of v and x as $t \rightarrow \infty$. **2**

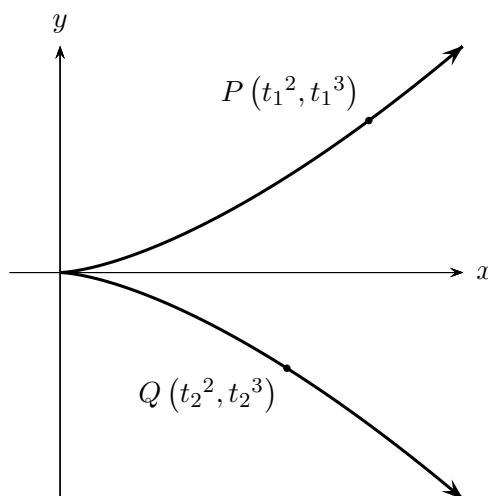
Examination continues overleaf...

Question 16 (15 Marks)

Commence a NEW booklet.

Marks

- (a) Let $f(x) = \frac{1-x}{x}$. On separate one-third page diagrams, sketch the following, clearly showing any asymptotes.
- i. $y = f(x)$ 1
 - ii. $y = f(|x|)$ 2
 - iii. $y = e^{f(x)}$ 2
 - iv. $y^2 = f(x)$. In addition, briefly describe the behaviour of the curve at $x = 1$. 3
- (b) Consider the curve defined parametrically by $\begin{cases} x = t^2 \\ y = t^3 \end{cases}$.



Let $P(t_1^2, t_1^3)$ and $Q(t_2^2, t_2^3)$ be two distinct points on the curve.

- i. Express the equation of the curve in terms of x and y only. 1
- ii. Show that the equation of the chord PQ is given by 2

$$(t_1 + t_2)y - (t_1^2 + t_1t_2 + t_2^2)x + t_1^2t_2^2 = 0$$

- iii. Hence or otherwise, show that the equation of the tangent to the curve at a point corresponding to t , where $t \neq 0$, is given by 1

$$2y - 3tx + t^3 = 0$$

- iv. Let $R(x_0, y_0)$ be a point in the plane such that $x_0^3 > y_0^2 > 0$. 4

Prove that there are exactly three tangents from R to the curve.

(Hint: differentiate the expression given in part (iii))

End of paper.

2017 Mathematics Extension 2 HSC Course Assessment Task 4

STUDENT SELF REFLECTION

1. In hindsight, did I do the best I can? Why or why not?

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- Q15(a)(b) - Volumes

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2. Which topics did I need more help with, and what parts specifically?

- Q5, 8, 10, 11 - Integration

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- Q15(c) - Mechanics (Resisted Motion)

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- Q1, 2, 9, 16(a) - Graphs

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- Q6, 7, 12 - Complex Numbers

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3. What other parts from the feedback session can I take away to refine my solutions for future reference?

- Q3, 13, 16(b) - Polynomials

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- Q14 - Conics

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Sample Band 6/E4 Responses

Section I

1. (C) 2. (C) 3. (B) 4. (A) 5. (B)
6. (D) 7. (B) 8. (C) 9. (A) 10. (D)

Section II

Question 11 (Bhamra)

(a) (4 marks)

- ✓ [1] for the correct expression by applying integration by parts the first time.
- ✓ [1] for correctly applying integration by parts, the second time.
- ✓ [1] for observing the original in the second application of integration by parts
- ✓ [1] for final answer.

$$\text{Let } I = \int e^{-2x} \cos x \, dx.$$

$$\left| \begin{array}{ll} u = e^{-2x} & dv = \cos x \\ du = -2e^{-2x} & v = \sin x \end{array} \right|$$

$$I = uv - \int v \, du$$

$$= e^{-2x} \sin x - \int -2e^{-2x} \sin x \, dx$$

$$= e^{-2x} \sin x + 2 \int e^{-2x} \sin x \, dx$$

For the rightmost integral,

$$\left| \begin{array}{ll} u = e^{-2x} & dv = \sin x \\ du = -2e^{-2x} & v = -\cos x \end{array} \right|$$

$$\therefore I = e^{-2x} \sin x + 2 \left(-e^{-2x} \cos x - 2 \int e^{-2x} \cos x \, dx \right) \quad (2 \text{ marks})$$

$$= e^{-2x} \sin x - 2e^{-2x} \cos x - 4I + C_1$$

$$5I = e^{-2x} \sin x - 2e^{-2x} \cos x + C_1$$

$$I = \frac{1}{5} e^{-2x} (\sin x - 2 \cos x) + C$$

Alternatively, let $u = \cos x$ and $dv = e^{-2x}$. Correct application should result in the same primitive.

(b) (3 marks)

- ✓ [1] for correctly manipulation of integrand.

- ✓ [1] for correct resolution into index form.
- ✓ [1] for final answer.

$$\begin{aligned} \int \frac{x}{\sqrt{1-x}} \, dx &= - \int \left(\frac{1-x-1}{\sqrt{1-x}} \right) \, dx \\ &= - \int \left(\sqrt{1-x} - \frac{1}{\sqrt{1-x}} \right) \, dx \\ &= - \int \left((1-x)^{\frac{1}{2}} + (1-x)^{-\frac{1}{2}} \right) \, dx \\ &= \frac{2}{3} (1-x)^{\frac{3}{2}} + 2(1-x)^{\frac{1}{2}} + C \end{aligned}$$

Alternatively, make the substitution $u^2 = 1-x$, or $u = 1-x$.

(c) i. (2 marks)

- ✓ [1] for substitution transformation.
- ✓ [1] for correct proof.

Letting $x = a - u$,

$$\left| \begin{array}{ll} x = a - u & x = 0, u = a \\ dx = -du & x = a, u = 0 \end{array} \right|$$

$$\therefore \int_0^a f(x) \, dx = \int_{u=a}^{u=0} f(a-u) (-du)$$

$$= \int_0^a f(a-u) \, du$$

$$= \int_0^a f(a-x) \, dx$$

by interchanging u for x .

- ✓ [1] for integrand transformation.
- ✓ [1] for final answer.

$$\begin{aligned} \int_0^1 x(1-x)^{2017} \, dx &= \int_0^1 x^{2017} (1-x) \, dx \\ &= \int_0^1 (x^{2017} - x^{2018}) \, dx \\ &= \left[\frac{x^{2018}}{2018} - \frac{x^{2019}}{2019} \right]_0^1 \\ &= \frac{1}{2018} - \frac{1}{2019} \\ &= \frac{1}{(2018)(2019)} \end{aligned}$$

(d) (4 marks)

- ✓ [1] for correct t transformation, including corresponding differential.
- ✓ [1] for simplification of the new denominator in terms of t .
- ✓ [1] for transformation into an integrand leading to the inverse trig integral.
- ✓ [1] for final answer.

Letting $t = \tan \frac{x}{2}$, then $x = 2 \tan^{-1} t$:

$$\begin{aligned}
 & \left| \begin{array}{ll} x = 2 \tan^{-1} t & x = 0, t = 0 \\ dx = \frac{2}{1+t^2} dt & x = \frac{\pi}{2}, t = 1 \end{array} \right. \\
 & \int_0^{\frac{\pi}{2}} \frac{1}{3 - \cos x - 2 \sin x} dx \\
 &= \int_{t=0}^{t=1} \frac{1}{3 - \frac{1-t^2}{1+t^2} - 2 \left(\frac{2t}{1+t^2} \right)} \cdot \frac{2}{1+t^2} dt \\
 &= \int_0^1 \frac{2 dt}{3 + 3t^2 - 1 + t^2 - 4t} \\
 &= \int_0^1 \frac{2 dt}{4t^2 - 4t + 2} \\
 &= \int_0^1 \frac{2 dt}{4t^2 - 4t + \textcolor{red}{1} + \textcolor{red}{1}} \\
 &= \int_0^1 \frac{2 dt}{(2t-1)^2 + 1} \\
 &= [\tan^{-1}(2t-1)]_0^1 \\
 &= \tan^{-1}(1) - \tan^{-1}(-1) = \frac{\pi}{2}
 \end{aligned}$$

ii. (2 marks)

- z^7 by De Moivre's Theorem:

$$\begin{aligned}
 z^7 &= 2^7 \left(\cos \frac{7\pi}{3} + i \sin \frac{7\pi}{3} \right) \\
 &= 128 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)
 \end{aligned}$$

- $64z$:

$$\begin{aligned}
 64z &= 64 \times 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) \\
 &= 128 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)
 \end{aligned}$$

Hence $z^7 - 64z = 0$.

(b) (4 marks)

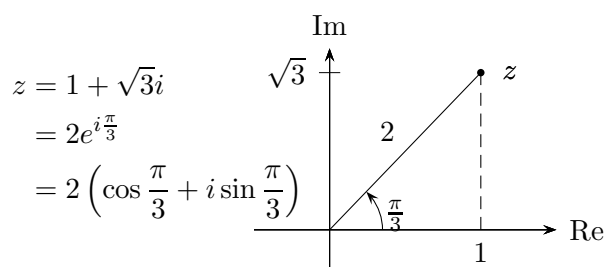
- ✓ [1] for formation of simultaneous equations in terms of a and b .
- ✓ [1] for solutions to $\sqrt{-8+6i}$
- ✓ [1] for applying quadratic formula to complex quadratic.
- ✓ [1] for both final solutions.
- Evaluate $\sqrt{6i-8} = \sqrt{-8+6i}$. Letting $z^2 = -8+6i$ where $z = a+ib$ and then comparing real and imaginary parts:

$$\begin{aligned}
 (a+ib)^2 &\equiv -8+6i \\
 a^2 - b^2 + i(2ab) &\equiv -8+6i \\
 \begin{cases} a^2 - b^2 = -8 & (1) \\ ab = 3 & (2) \end{cases}
 \end{aligned}$$

Question 12 (Bhamra)

Substituting (2) into (1):

(a) i. (1 mark)



$$\begin{aligned}
 a^2 - \left(\frac{3}{a} \right)^2 &= -8 \\
 a^4 - 9 &= -8a^2 \\
 a^4 + 8a^2 - 9 &= 0 \\
 (a^2 + 9)(a^2 - 1) &= 0 \\
 \therefore a &= \pm 1 & b &= \pm 3 \\
 \therefore \sqrt{-8+6i} &= \pm(1+3i)
 \end{aligned}$$

- Solving $2z^2 - (3+i)z + 2 = 0$ via the quadratic formula:

$$\begin{aligned}
 z &= \frac{(3+i) \pm \sqrt{(3+i)^2 - 16}}{4} \\
 &= \frac{(3+i) \pm \sqrt{9+6i-1-16}}{4} \\
 &= \frac{(3+i) \pm \sqrt{-8+6i}}{4} \\
 &= \frac{(3+i) \pm (1+3i)}{4} \\
 z_1 &= \frac{3+i+1+3i}{4} = 1+i \\
 z_2 &= \frac{3+i-1-3i}{4} = \frac{1}{2} - \frac{1}{2}i
 \end{aligned}$$

- (c) (2 marks) Let $z = a + ib$,

$$\begin{aligned}
 2\bar{z} - iz &= 1 + 4i \\
 2(a - ib) - i(a + ib) &= 1 + 4i \\
 2a - 2ib - ai + b &= 1 + 4i \\
 2a + b - i(a + 2b) &= 1 + 4i
 \end{aligned}$$

Equating real and imaginary parts,

$$\begin{cases} 2a + b = 1 & (1) \\ -a - 2b = 4 & (2) \end{cases}$$

Multiplying (1) by 2, and adding to (2):

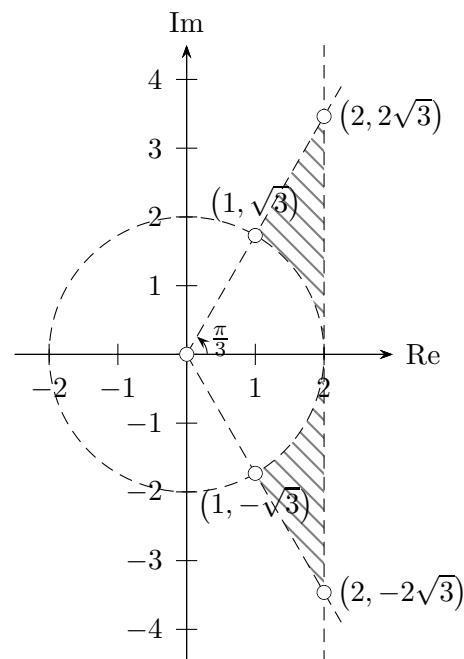
$$\begin{aligned}
 &\begin{cases} 4a + 2b = 2 & (1) \times 2 \\ -a - 2b = 4 & (2) \end{cases} \\
 &3a = 6 \\
 &a = 2
 \end{aligned}$$

Substitute into (2):

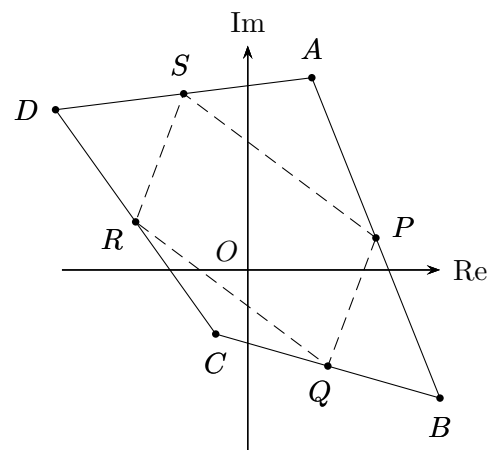
$$\begin{aligned}
 -2 - 2b &= 4 \\
 -2b &= 6 \\
 b &= -3 \\
 \therefore z &= 2 - 3i
 \end{aligned}$$

- (d) (3 marks)

- $|\arg(z)| < \frac{\pi}{3}$ draw straight lines angled at $\pm \frac{\pi}{3}$ to positive x axis
- $z + \bar{z} < 4$: results in region $2x < 4$
- $|z| > 2$: region outside the circle of radius 2, centre origin



- (e) i. (2 marks)



$$\bullet \quad \overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP}.$$

$$\text{Hence } \overrightarrow{OP} = \overrightarrow{OA} + \frac{1}{2}\overrightarrow{AB}:$$

$$\begin{aligned}
 \overrightarrow{OP} &\Rightarrow a + \frac{1}{2}(b - a) \\
 &= \frac{1}{2}(a + b)
 \end{aligned}$$

- Similarly, $\overrightarrow{OR}$ is represented by

$$\frac{1}{2}(c + d)$$

- As M is the midpoint of PR ,

$$\begin{aligned}\overrightarrow{OM} &= \frac{1}{2} (\overrightarrow{OP} + \overrightarrow{OR}) \\ &\Rightarrow \frac{1}{2} \left(\frac{1}{2}(a+b) + \frac{1}{2}(c+d) \right) \\ &= \frac{1}{4}(a+b+c+d)\end{aligned}$$

- N is the midpoint of QS , and $\overrightarrow{ON}$ represents

$$\begin{aligned}\overrightarrow{ON} &= \frac{1}{2} (\overrightarrow{OQ} + \overrightarrow{OS}) \\ &\Rightarrow \frac{1}{2} \left(\frac{1}{2}(b+c) + \frac{1}{2}(a+d) \right) \\ &= \frac{1}{4}(a+b+c+d)\end{aligned}$$

- ii. (1 mark) As M and N are coincident, the diagonals of $PQRS$ bisect each other. $PQRS$ is a parallelogram.

Question 13 (Sharma)

- (a) i. (1 mark)

$$x^3 + 0x^2 + px + q = 0$$

- (b) i. (3 marks)

✓ [1] for each correct constant.

With roots α , β and γ :

$$\begin{aligned}(\alpha + \beta + \gamma)^2 &= \alpha^2 + \beta^2 + \gamma^2 + 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\ \therefore \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)\end{aligned}$$

Finding the elementary symmetric functions,

$$\begin{aligned}\alpha + \beta + \gamma &= 0 \\ \alpha\beta + \beta\gamma + \alpha\gamma &= \frac{p}{1} = p \\ \alpha\beta\gamma &= -\frac{q}{1} = -q \\ \therefore \alpha^2 + \beta^2 + \gamma^2 &= (0)^2 - 2(p) = -2p\end{aligned}$$

- ii. (3 marks)

- ✓ [1] for identifying new roots of the new equation.
 ✓ [1] for correct polynomial with roots α^2 , β^2 and γ^2 .
 ✓ [1] for correct polynomial with roots $\frac{\alpha^2}{-q}$ etc.

$$\frac{\alpha}{\beta\gamma} = \frac{\alpha^2}{\alpha\beta\gamma} \quad \frac{\beta}{\alpha\gamma} = \frac{\beta^2}{\alpha\beta\gamma} \quad \frac{\gamma}{\alpha\beta} = \frac{\gamma^2}{\alpha\beta\gamma}$$

Hence, new equation will have roots

$$\frac{\alpha^2}{-q}, \frac{\beta^2}{-q}, \frac{\gamma^2}{-q}$$

In $x^3 + px + q = 0$, let $x \mapsto \sqrt{x}$, which produces roots α^2 , β^2 and γ^2 :

$$\begin{aligned}(\sqrt{x})^3 + p(\sqrt{x}) + q &= 0 \\ x\sqrt{x} + p\sqrt{x} + q &= 0 \\ \sqrt{x}(x+p) &= -q \\ x(x+p)^2 &= q^2 \\ x(x^2 + 2px + p^2) &= q^2 \\ x^3 + 2px^2 + p^2x - q^2 &= 0\end{aligned}$$

Now let $x \mapsto -qx$ to produce the required roots:

$$\begin{aligned}(-qx)^3 + 2p(-qx)^2 + p^2(-qx) - q^2 &= 0 \\ -q^3x^3 + 2pq^2x^2 - p^2qx - q^2 &= 0 \\ \therefore q^2x^3 - 2pqx^2 + p^2x + q &= 0\end{aligned}$$

$$\begin{aligned}\frac{4x^2 - 5x - 7}{(x-1)(x^2+x+2)} &\equiv \frac{A}{x-1} + \frac{Bx+C}{x^2+x+2} \\ 4x^2 - 5x - 7 &\equiv A(x^2+x+2) + (Bx+C)(x-1)\end{aligned}$$

- Let $x = 1$,

$$\begin{aligned}4 - 5 - 7 &\equiv A(1 + 1 + 2) + 0 \\ 4A &= -8 \\ A &= -2\end{aligned}$$

- Let $x = 0$:

$$\begin{aligned}-7 &\equiv -2(0 + 0 + 2) + (B(0) + C)(0 - 1) \\ -7 &\equiv -4 - C \\ C &= 3\end{aligned}$$

- Let $x = 2$,

$$\begin{aligned}4(2^2) - 5(2) - 7 &= 16 - 10 - 7 = -16 + 2B + 3 \\ &\equiv -2(2^2 + 2 + 2) + (2B + 3)(2 - 1) \\ 16 - 10 - 7 &\equiv -16 + 2B + 3 \\ 2B &= 12 \\ B &= 6\end{aligned}$$

$$\therefore \frac{4x^2 - 5x - 7}{(x-1)(x^2+x+2)} \equiv \frac{-2}{x-1} + \frac{6x+3}{x^2+x+2}$$

ii. (2 marks)

✓ [1] for correct integral.

✓ [1] for correct final answer.

$$\begin{aligned} & \int_{-1}^0 \frac{4x^2 - 5x - 7}{(x-1)(x^2+x+2)} dx \\ &= \int_{-1}^0 \left(-\frac{2}{x-1} + \frac{3(2x+1)}{x^2+x+2} \right) dx \\ &= -2 [\ln(x-1)]_{-1}^0 + 3 [\ln(x^2+x+2)]_{-1}^0 \end{aligned}$$

By symmetry,

$$\int_{-1}^0 -\frac{2}{x-1} dx = \int_2^3 \frac{2}{x-1} dx$$

(Alternatively, use absolute values inside the logarithm)

$$\begin{aligned} & \int_{-1}^0 \left(-\frac{2}{x-1} + \frac{3(2x+1)}{x^2+x+2} \right) dx \\ &= 2 [\ln(x-1)]_2^3 + 3 [\ln(x^2+x+2)]_{-1}^0 \\ &= 2 \ln 2 + 3 (\ln 2 - \ln(1-1+2)) \\ &= 2 \ln 2 \end{aligned}$$

(c) i. (2 marks)

$$\begin{aligned} & \cos 3A \\ &= \cos(A+2A) \\ &= \cos A \cos 2A - \sin A \sin 2A \\ &= \cos A (2\cos^2 A - 1) - \sin A (2\sin A \cos A) \\ &= 2\cos^3 A - \cos A - 2(1 - \cos^2 A) \cos A \\ &= 2\cos^3 A - \cos A - 2\cos A + 2\cos^3 A \\ &= 4\cos^3 A - 3\cos A \\ &\therefore \frac{1}{4} \cos 3A = \cos^3 A - \frac{3}{4} \cos A \end{aligned}$$

Alternatively, use De Moivre's theorem to expand and equate real parts to obtain required result.

ii. (2 marks)

✓ [1] for $x^3 - 6x$ in terms of $\cos A$.

✓ [1] for correctly showing required result.

$$x = 2\sqrt{2} \cos A$$

$$x^3 = 16\sqrt{2} \cos^3 A$$

$$\begin{aligned} \therefore x^3 - 6x &= 16\sqrt{2} \cos^3 A - 6(2\sqrt{2} \cos A) \\ &= 16\sqrt{2} \cos^3 A - 12\sqrt{2} \cos A \\ &= 4\sqrt{2} (4\cos^3 A - 3\cos A) \\ &= 4\sqrt{2} \cos 3A \end{aligned}$$

$$\text{If } \cos 3A = -\frac{1}{2\sqrt{2}},$$

$$\begin{aligned} x^3 - 6x &= 4\sqrt{2} \left(-\frac{1}{2\sqrt{2}} \right) \\ &= -2 \end{aligned}$$

Hence $x^3 - 6x = -2$ has a solution of $x = 2\sqrt{2} \cos A$ provided $\cos 3A = -\frac{1}{2\sqrt{2}}$.

iii. (2 marks)

✓ [1] for first quadrant related angle.

✓ [1] for all three roots.

$$\begin{aligned} \cos 3A &= -\frac{1}{2\sqrt{2}} \\ 3A &= 1.93 \dots \end{aligned}$$

$3A = 1.93 \dots$ lies in the 2nd quadrant. The first quadrant (related) angle is

$$\pi - 1.93 \dots = 1.209 \dots$$

Hence,

$$3A = \pi - 1.209 \dots \quad (2\text{nd quadrant})$$

$$\text{or } \pi + 1.209 \dots \quad (3\text{rd quadrant})$$

$$\text{or } 3\pi - 1.209 \dots \quad (6\text{th quadrant})$$

$$A = \frac{\pi - 1.209 \dots}{3}, \frac{\pi + 1.209 \dots}{3}, \frac{3\pi - 1.209 \dots}{3}$$

$$\therefore x = 2\sqrt{2} \cos \left(\frac{\pi - 1.209 \dots}{3} \right) \approx 2.2618 \dots$$

$$\text{or } 2\sqrt{2} \cos \left(\frac{\pi + 1.209 \dots}{3} \right) \approx 0.3399 \dots$$

$$\text{or } 2\sqrt{2} \cos \left(\frac{3\pi - 1.209 \dots}{3} \right) \approx -2.6017 \dots$$

Question 14 (Sharma)

(a) i. (α) (1 mark)

$$\frac{x^2}{3+p} + \frac{y^2}{8+p} = 1$$

If the conic is an ellipse, then both of $3+p > 0$ and $8+p > 0$:

$$\begin{array}{ll} 3+p > 0 & 8+p > 0 \\ p > -3 & p > -8 \end{array}$$

Hence $p > -3$ for both inequalities to be satisfied.

(α) (2 marks)

✓ [1] for the first condition $p > -3$ or $p < -8$, and explanation of which represents the most restrictive condition.

✓ [1] for rejecting the invalid solution.

If the conic is a hyperbola, then either of the following pairs of inequalities must hold true simultaneously:

$$- 3+p > 0 \text{ and } 8+p < 0$$

$$p > -3 \text{ and } p < -8$$

which is not possible to hold simultaneously.

$$- 3+p < 0 \text{ and } 8+p > 0$$

$$p < -3 \text{ and } p > -8$$

Hence $-8 < p < -3$.

ii. (2 marks)

✓ [-1] for each new error.

$$p = -4: \quad \frac{x^2}{3-4} + \frac{y^2}{8-4} = 1$$

$$\frac{x^2}{-1} + \frac{y^2}{4} = 1$$

$$\therefore \frac{y^2}{4} - x^2 = 1$$

Applying relationship for eccentricity:

$$b^2 = a^2 (e^2 - 1)$$

$$1 = 4 (e^2 - 1)$$

$$e^2 = \frac{5}{4}$$

$$e = \frac{\sqrt{5}}{2}$$

Finding the foci with $a = 2$ and $e = \frac{\sqrt{5}}{2}$:

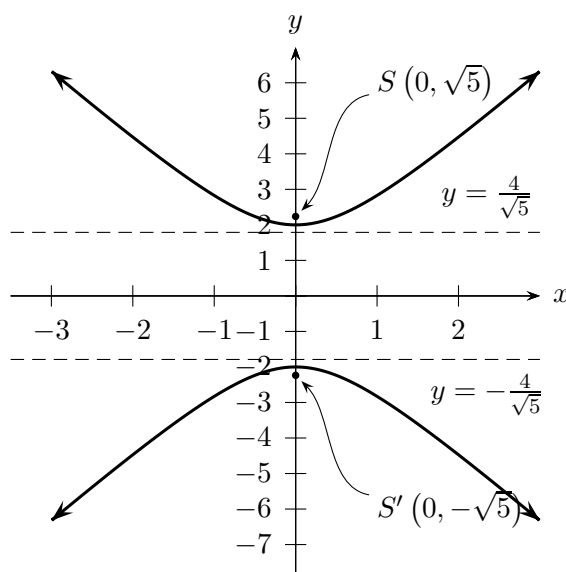
$$S(0, \pm ae) = (0, \pm\sqrt{5})$$

Finding the directrices:

$$y = \pm \frac{a}{e} = \pm \frac{2}{\frac{\sqrt{5}}{2}} = \pm \frac{4}{\sqrt{5}}$$

iii. (2 marks)

✓ [-1] for each new error.



(b) i. (2 marks)

$$y = \frac{1}{x}$$

$$\frac{dy}{dx} = -x^{-2} \Big|_{x=t} = -\frac{1}{t^2}$$

Equation of the tangent at $(t, \frac{1}{t})$ via the point-gradient formula:

$$\begin{aligned} \frac{y - \frac{1}{t}}{x - t} &= -\frac{1}{t^2} \\ t^2 y - t &= -x + t \\ x + t^2 y &= 2t \end{aligned}$$

ii. (3 marks)

✓ [1] for correct equation of the normal at $(t, \frac{1}{t})$.

- ✓ [1] for correct coordinates of A and B . (c) (3 marks)

- ✓ [1] for required proof.

- The equation of the normal at $(t, \frac{1}{t})$ via the point-gradient formula:

$$\begin{aligned}\frac{y - \frac{1}{t}}{x - t} &= t^2 \\ y - \frac{1}{t} &= t^2x - t^3 \\ ty - 1 &= t^3x - t^4 \\ t^3x - ty &= t^4 - 1\end{aligned}$$

- At $x = 0$, the tangent is at

$$\begin{aligned}0 + t^2y &= 2t \\ y &= \frac{2}{t} \\ \therefore A &\left(0, \frac{2}{t}\right)\end{aligned}$$

- At $y = 0$, the normal is at

$$\begin{aligned}t^3x - 0 &= t^4 - 1 \\ x &= \frac{t^4 - 1}{t^3} \\ \therefore B &\left(\frac{t^4 - 1}{t^3}, 0\right)\end{aligned}$$

- The midpoint of AB :

$$\begin{aligned}\left(\frac{\frac{t^4-1}{t^3} + 0}{2}, \frac{\frac{2}{t} + 0}{2}\right) &= \left(\frac{t^4 - 1}{2t^3}, \frac{1}{t}\right) \\ \therefore x_M &= \frac{t^4 - 1}{2t^3} \quad y_M = \frac{1}{t}\end{aligned}$$

Substituting $t = \frac{1}{y}$ into the x equation:

$$\begin{aligned}x &= \frac{\left(\frac{1}{y}\right)^4 - 1}{2\left(\frac{1}{y}\right)^3} = \frac{\frac{1}{y^4} - 1}{\frac{2}{y^3}} \times \frac{y^4}{y^4} \\ &= \frac{1 - y^4}{2y}\end{aligned}$$

- ✓ [1] for obtaining m_{PQ} .

- ✓ [1] for using the point gradient formula for the equation of PQ .

- ✓ [1] for final solution.

$$\begin{cases} x_P = a \cos \alpha & x_Q = a \cos \beta \\ y_P = b \sin \alpha & y_Q = b \sin \beta \end{cases}$$

The gradient of the chord PQ :

$$m_{PQ} = \frac{b \sin \beta - b \sin \alpha}{a \cos \beta - a \cos \alpha} = \frac{b(\sin \beta - \sin \alpha)}{a(\cos \beta - \cos \alpha)}$$

Using the point-gradient formula, the equation of PQ is:

$$\frac{y - b \sin \alpha}{x - a \cos \alpha} = \frac{b(\sin \beta - \sin \alpha)}{a(\cos \beta - \cos \alpha)}$$

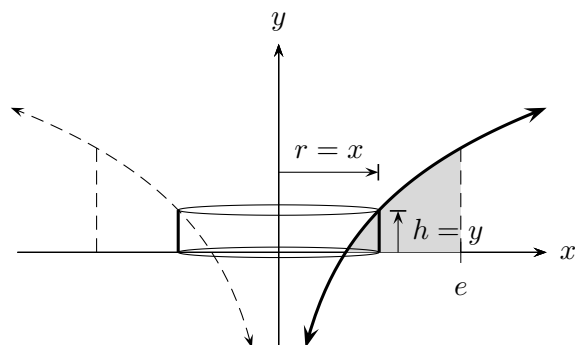
As the focal chord passes through $(ae, 0)$:

$$\begin{aligned}\frac{-b \sin \alpha}{ae - a \cos \alpha} &= \frac{b(\sin \beta - \sin \alpha)}{a(\cos \beta - \cos \alpha)} \\ -\sin \alpha(\cos \beta - \cos \alpha) &= (e - \cos \alpha)(\sin \beta - \sin \alpha) \\ e &= \frac{-\sin \alpha \cos \beta + \sin \alpha \cos \alpha}{\sin \beta - \sin \alpha} + \cos \alpha \\ &= \frac{-\sin \alpha \cos \beta + \sin \alpha \cos \alpha + \sin \beta \cos \alpha - \sin \alpha \cos \alpha}{\sin \beta - \sin \alpha} \\ &= \frac{\sin \beta \cos \alpha - \sin \alpha \cos \beta}{\sin \beta - \sin \alpha} \\ &= \frac{\sin(\beta - \alpha)}{\sin \beta - \sin \alpha}\end{aligned}$$

Question 15 (Lam)

(a) (3 marks)

- ✓ [1] for the correct working, leading to the curved surface area of the cylinder.
- ✓ [1] for the applying integration by parts.
- ✓ [1] for the final answer.



- Surface area of cylinder: $SA = 2\pi rh$
- Radius of each individual cylinder:

$$r = x$$

- Height of each individual cylinder:

$$h = y$$

- Hence,

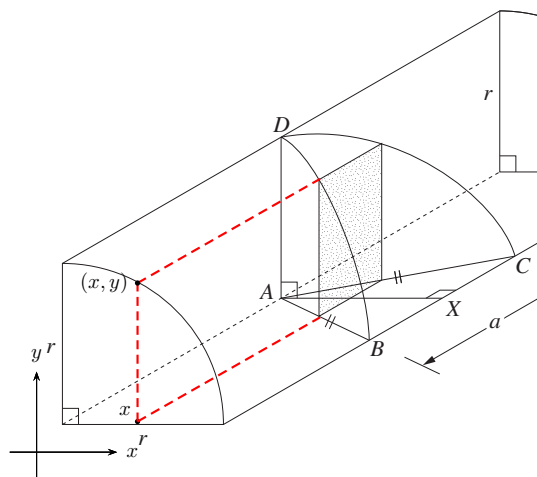
$$\begin{aligned}\delta V &= \sum A \delta x \\ &= \sum 2\pi xy \delta x \\ &= 2\pi \sum x \ln x \delta x\end{aligned}$$

Taking $\delta x \rightarrow 0$:

$$\begin{aligned}V &= 2\pi \lim_{\delta x \rightarrow 0} \sum_{x=1}^{x=e} x \ln x \delta x \\ &= 2\pi \int_1^e x \ln x \, dx \\ &\quad \left| \begin{array}{l} u = \ln x \quad dv = x \\ du = \frac{1}{x} \quad v = \frac{1}{2}x^2 \end{array} \right. \\ \therefore V &= 2\pi \left(\left[\frac{1}{2}x^2 \ln x \right]_1^e - \int_1^e \frac{1}{2}x^2 \cdot \frac{1}{x} \, dx \right) \\ &= 2\pi \left(\frac{1}{2}e^2 - \frac{1}{2} \left[\frac{1}{2}x^2 \right]_1^e \right) \\ &= 2\pi \left(\frac{1}{2}e^2 - \frac{1}{4}e^2 + \frac{1}{4} \right) \\ &= \frac{\pi}{2} (e^2 + 1)\end{aligned}$$

(b) (3 marks)

- ✓ [1] for similarity ratio relationship in $\triangle ABC$.
- ✓ [1] for correct area expression.
- ✓ [1] for final answer.

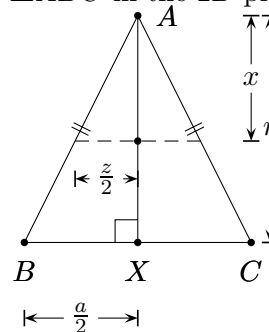


- From the diagram: at a point x along the horizontal, the corresponding vertical value is y . As it is a quarter circle,

$$y = \sqrt{r^2 - x^2}$$

which becomes the height of the rectangle.

- Drawing $\triangle ABC$ in the 2D plane:



From similar triangles,

$$\begin{aligned}\frac{r}{\frac{a}{2}} &= \frac{x}{\frac{z}{2}} \\ \therefore \frac{2r}{a} &= \frac{2x}{z} \\ z &= \frac{ax}{r}\end{aligned}$$

- From the area of the rectangles,

$$A = bh = zy = \frac{ax}{r} \sqrt{r^2 - x^2}$$

the volume is the sum of these areas:

$$\begin{aligned} V &= \lim_{\delta x \rightarrow 0} \sum_{x=0}^{x=r} A \delta x = \int_0^r \frac{ax}{r} \sqrt{r^2 - x^2} dx \\ &= \frac{a}{r} \times \left(-\frac{1}{2}\right) \int_0^r -2x (r^2 - x^2)^{\frac{1}{2}} dx \\ &= -\frac{a}{2r} \left[\frac{2}{3} (r^2 - x^2)^{\frac{3}{2}} \right]_0^r \\ &= -\frac{a}{2r} \left(0 - (r^2)^{\frac{3}{2}} \right) = \frac{ar^2}{3} \end{aligned}$$

(c) i. (2 marks)

- ✓ [1] for separation of variables to obtain $\tan^{-1} x = -t + C_1$.
- ✓ [1] for final required result.

$$F = m\ddot{x} = -(v + v^3)$$

As $m = 1$,

$$\begin{aligned} \ddot{x} &= v \frac{dv}{dx} = -(v + v^3) \\ \frac{1}{1 + v^2} dv &= -dx \end{aligned}$$

Integrating both sides,

$$\tan^{-1} v = -x + C_1$$

When $t = 0$, $v = Q$, $x = 0$.

$$\begin{aligned} \therefore C_1 &= \tan^{-1} Q \\ \therefore x &= \tan^{-1} Q - \tan^{-1} v \end{aligned}$$

Letting

$$\begin{aligned} \alpha &= \tan^{-1} Q & \beta &= \tan^{-1} v \\ \tan \alpha &= Q & \tan \beta &= v \\ \therefore \tan x &= \tan(\alpha - \beta) \\ &= \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta} \\ &= \frac{Q - v}{1 + Qv} \\ \therefore x &= \tan^{-1} \left(\frac{Q - v}{1 + Qv} \right) \end{aligned}$$

ii. (3 marks)

- ✓ [1] for correct constants in the partial fraction decomposition.
- ✓ [1] for obtaining the correct primitive.
- ✓ [1] for final required result.

$$\begin{aligned} \frac{dv}{dt} &= -v - v^3 \\ \frac{dv}{v(1 + v^2)} &= -dt \end{aligned}$$

Applying the partial fraction decomposition,

$$\frac{1}{v(1 + v^2)} \equiv \frac{A}{v} + \frac{Bv + C}{1 + v^2}$$

$\times v(1 + v^2)$

$$1 \equiv A(1 + v^2) + v(Bv + C)$$

- When $v = 0$, $A = 1$ by inspection.
- When $v = 1$,

$$\begin{aligned} 1 &\equiv 1(1 + 1) + 1(B + C) \\ 1 &\equiv 2 + B + C \\ \therefore B + C &= -1 \end{aligned}$$

- When $v = 2$,

$$\begin{aligned} 1 &\equiv 1(1 + 4) + 2(2B + C) \\ 1 &\equiv 5 + 4B + 2C \\ 4B + 2C &= -4 \\ 2B + C &= -2 \end{aligned}$$

- Solving simultaneously,

$$\begin{cases} B + C = -1 & (1) \\ 2B + C = -2 & (2) \end{cases}$$

(2) - (1):

$$\begin{aligned} B &= -2 - (-1) = -1 \\ \therefore C &= 0 \end{aligned}$$

$$\begin{aligned} \int \frac{dv}{v(1 + v^2)} &= \int \left(\frac{1}{v} - \frac{v}{1 + v^2} \right) dv \\ &= \int -dt \\ \ln v - \frac{1}{2} \ln(1 + v^2) &= -t + C_2 \end{aligned}$$

When $t = 0$, $v = Q$:

Question 16 (Lam)

$$C_2 = \ln Q - \frac{1}{2} \ln(1 + Q^2)$$

$$\ln v - \frac{1}{2} \ln(1 + v^2) \quad (\text{a}) \quad \text{i. (1 mark)}$$

$$= -t + \ln Q - \frac{1}{2} \ln(1 + Q^2)$$

$$\begin{aligned} t &= \ln Q - \ln v \\ &\quad + \frac{1}{2} (\ln(1 + v^2) - \ln(1 + Q^2)) \\ &= \ln \frac{Q}{v} + \frac{1}{2} \ln \left(\frac{1 + v^2}{1 + Q^2} \right) \\ &= \frac{1}{2} \ln \frac{Q^2}{v^2} + \frac{1}{2} \ln \left(\frac{1 + v^2}{1 + Q^2} \right) \\ &= \frac{1}{2} \ln \left(\frac{Q^2 (1 + v^2)}{v^2 (1 + Q^2)} \right) \end{aligned}$$

iii. (2 marks)

- ✓ [1] for e^{2t} as the subject.
- ✓ [1] for final result required.

$$t = \frac{1}{2} \ln \left(\frac{Q^2 (1 + v^2)}{v^2 (1 + Q^2)} \right)$$

$$2t = \ln \left(\frac{Q^2 (1 + v^2)}{v^2 (1 + Q^2)} \right)$$

$$e^{2t} = \frac{Q^2 (1 + v^2)}{v^2 (1 + Q^2)}$$

$$\begin{aligned} e^{2t} v^2 + e^{2t} Q^2 v^2 &= Q^2 + Q^2 v^2 \\ v^2 (e^{2t} + e^{2t} Q^2 - Q^2) &= Q^2 \end{aligned}$$

$$\begin{aligned} \therefore v^2 &= \frac{Q^2}{e^{2t} (1 + Q^2) - Q^2} \frac{\times e^{-2t}}{\times e^{-2t}} \\ &= \frac{Q^2 e^{-2t}}{(1 + Q^2) - Q^2 e^{-2t}} \end{aligned}$$

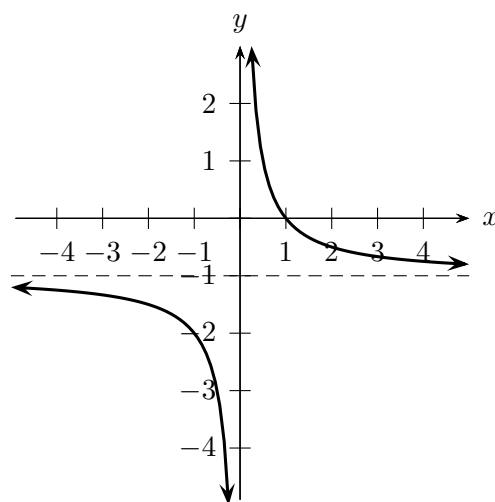
iv. (2 marks)

$$t \rightarrow \infty, e^{-2t} \rightarrow 0$$

$$\therefore v^2 \rightarrow \frac{0}{0 - Q^2} = 0$$

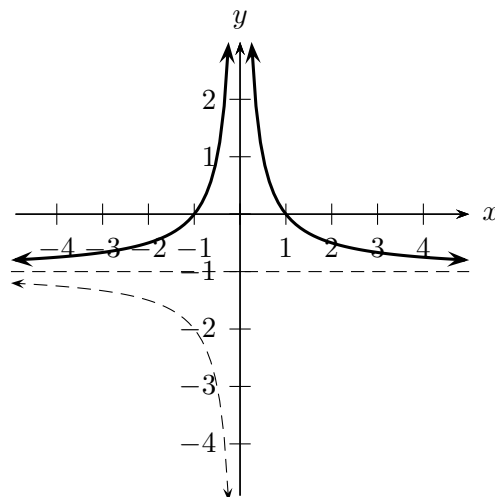
$$x \rightarrow \tan^{-1} \left(\frac{Q - 0}{1 + Q(0)} \right) = \tan^{-1} Q$$

$$\begin{aligned} y &= \frac{1-x}{x} = \frac{1}{x} - 1 \\ &= -1 + \frac{1}{x} \end{aligned}$$



ii. (2 marks)

$$y = f(|x|)$$



iii. (2 marks)

✓ [-1] for each new error/feature not shown. Features are:

- Shape (both branches)
- Open circle at $x = 0$ for negative branch
- y asymptote of $y = \frac{1}{e}$

$$y = e^{f(x)}$$

Asymptotes:

- $x \rightarrow \infty$:

$$y = f(x) = -1 + \frac{1}{x} \rightarrow -1^+$$

$$\therefore y = e^{f(x)} \rightarrow e^{-1} = \left(\frac{1}{e}\right)^+$$

- $x \rightarrow -\infty$:

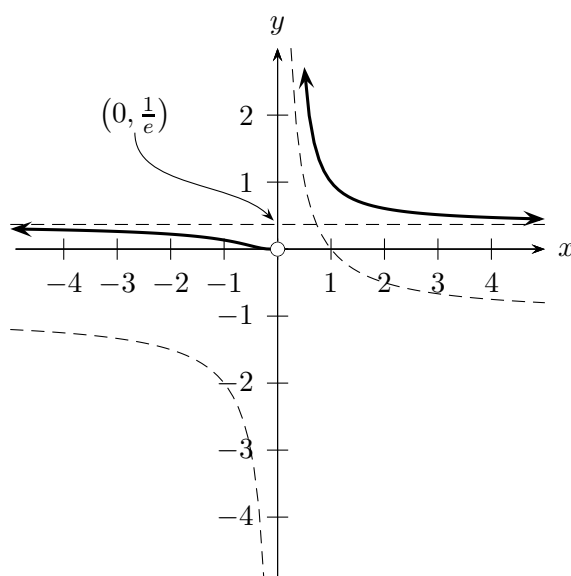
$$y = f(x) = -1 + \frac{1}{x} \rightarrow -1^-$$

$$\therefore y = e^{f(x)} \rightarrow e^{-1} = \left(\frac{1}{e}\right)^-$$

- $x \rightarrow 0^-$:

$$y = f(x) \rightarrow -\infty$$

$$\therefore y = e^{f(x)} \rightarrow 0^+$$



iv. (3 marks)

✓ [-1] for each new error/feature not shown. Features are:

- Shape (both branches)
- Smooth curve at $(1, 0)$
- x asymptote of $x = 0$

✓ [1] for description of the behaviour at $x = 1$, based on graph drawn.

$$y^2 = f(x)$$

- $y = \sqrt{f(x)}$ requires $f(x) \geq 0$:

$$\frac{1-x}{x} \geq 0$$

$$\frac{1}{x} - 1 \geq 0$$

$$\frac{1}{x} \geq 1$$

$$\therefore 0 < x \leq 1$$

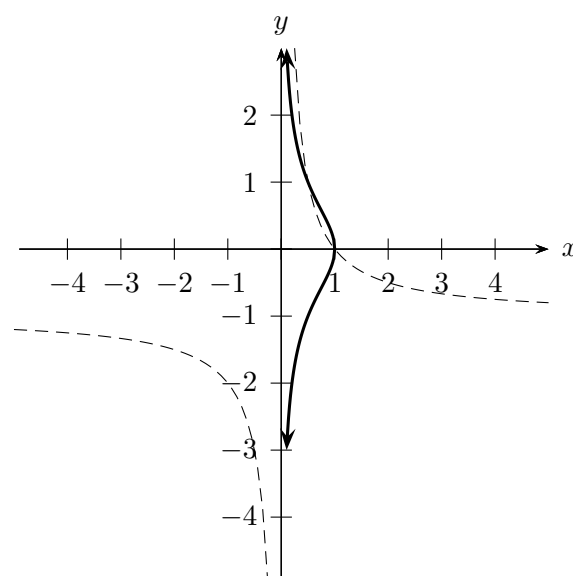
Hence the domain is $0 < x < 1$.

- Also, the other branch is

$$y = -\sqrt{f(x)}$$

Same domain, but below x axis.

- At $x = 1$, the curve reaches its maximum x value/at the upper limit of its domain and has a **vertical tangent**.



(b) i. (1 mark)

✓ [0] if $y = x\sqrt{x}$ (without $\pm$ symbol)

$$\begin{cases} x = t^2 & \Rightarrow x^3 = t^6 \\ y = t^3 & \Rightarrow y^2 = t^6 \end{cases}$$

Equating and removing the parameter,

$$y^2 = x^3$$

ii. (2 marks)

Finding the gradient of PQ :

$$\begin{aligned} m_{PQ} &= \frac{t_1^3 - t_2^3}{t_1^2 - t_2^2} \\ &= \frac{\cancel{(t_1 - t_2)}(t_1^2 + t_1t_2 + t_2^2)}{\cancel{(t_1 - t_2)}(t_1 + t_2)} \\ &= \frac{t_1^2 + t_1t_2 + t_2^2}{t_1 + t_2} \end{aligned}$$

Using the point-gradient formula,

$$\begin{aligned} \frac{y - t_1^3}{x - t_1^2} &= \frac{t_1^2 + t_1t_2 + t_2^2}{t_1 + t_2} \\ (y - t_1^3)(t_1 + t_2) &= (x - t_1^2)(t_1^2 + t_1t_2 + t_2^2) \\ y(t_1 + t_2) - \cancel{t_1^4} - \cancel{t_1^3t_2} &= x(t_1^2 + t_1t_2 + t_2^2) - \cancel{t_1^4} - \cancel{t_1^3t_2} - t_1^2t_2^2 \\ \therefore (t_1 + t_2)y - (t_1^2 + t_1t_2 + t_2^2)x + t_1^2t_2^2 &= 0 \end{aligned}$$

iii. (1 mark) When $t_1 = t_2$, the chord becomes a tangent.

$$\begin{aligned} 2t_1y - (t_1^2 + t_1^2 + t_1^2)x + t_1^4 &= 0 \\ 2t_1y - 3t_1^2x + t_1^4 &= 0 \\ \therefore 2y - 3tx + t^3 &= 0 \end{aligned}$$

iv. (4 marks)

- ✓ [1] for differentiating w.r.t. t and checking for stationary points.
- ✓ [1] for identifying cubic in terms of t will have three roots via fundamental theorem of algebra.
- ✓ [1] each for testing values of $f(\pm\sqrt{x_0})$.

✓ [1] for final reasoning why there are three real roots and therefore three unique tangents.

Tangent to any point $R(x_0, y_0)$, where $x_0^3 > y_0^2 > 0$ is

$$t^3 - 3tx_0 + 2y_0 = 0$$

Letting $f(t) = t^3 - 3tx_0 + 2y_0$. There are up to three unique solutions to this equation in terms of t . Need all three to be real and no complex conjugate pairs. Differentiating,

$$f'(t) = 3t^2 - 3x_0 = 3(t^2 - x_0)$$

Hence stationary points are located at $t = \pm\sqrt{x_0}$. Check actual $f(\sqrt{x_0})$ and $f(-\sqrt{x_0})$:

$$\begin{aligned} f(\sqrt{x_0}) &= (\sqrt{x_0})^3 - 3(\sqrt{x_0})x_0 + 2y_0 \\ &= -2x_0\sqrt{x_0} + 2y_0 \\ &= -2\sqrt{x_0^3} + 2y_0 \\ &= 2(y_0 - \sqrt{x_0^3}) \end{aligned}$$

As $x_0^3 > y_0^2 > 0$, then $\sqrt{x_0^3} > y_0$

$$\therefore f(\sqrt{x_0}) < 0$$

Testing $f(-\sqrt{x_0})$,

$$\begin{aligned} f(-\sqrt{x_0}) &= (-\sqrt{x_0})^3 - 3(-\sqrt{x_0})x_0 + 2y_0 \\ &= 2x_0\sqrt{x_0} + 2y_0 \\ &= 2\sqrt{x_0^3} + 2y_0 \\ &= 2(\sqrt{x_0^3} + y_0) \end{aligned}$$

As $x_0^3 > y_0^2 > 0$, then $\sqrt{x_0^3} > y_0$

$$\therefore f(-\sqrt{x_0}) > 0$$

In $f(t)$, one of the y values of the stationary points is negative and the other is the positive. Hence they are on opposing sides of the horizontal axis, and $t^3 - 3tx + y = 0$ always has three real solutions for all values of (x_0, y_0) where $x_0^3 > y_0^2 > 0$ and three unique tangents.